

Jenkins: Remote Job Triggering and Installation Guide

This document provides a comprehensive overview of how to trigger Jenkins jobs remotely and details the installation procedures for Jenkins on various platforms.

1. Triggering Jenkins Jobs Remotely

Jenkins offers several robust methods for initiating builds remotely, catering to different integration scenarios and security requirements. These methods enable automation and integration with external systems or scripts.

1.1. Using the Remote Access API (HTTP POST)

The Jenkins Remote Access API allows jobs to be triggered via HTTP POST requests. This is a common method for programmatic interaction with Jenkins [1].

- **Endpoint:** For jobs without parameters, the endpoint is typically `JENKINS_URL/job/JOBNAME/build`. For parameterized jobs, `JENKINS_URL/job/JOBNAME/buildWithParameters` is used [1].
- **Authentication:** Authentication is crucial for security. It can be achieved using a Jenkins user's password or, more securely, an **API Token**. API tokens are preferred as they can be revoked without changing the user's main password [2]. Authentication is often handled via Basic Authentication in the HTTP request header.

Bash

```
curl -u username:api_token -X POST JENKINS_URL/job/JOBNAME/build
```

- **CSRF Protection:** If Cross-Site Request Forgery (CSRF) protection is enabled in Jenkins (which is recommended for security), a **crumb** must be included in the request header (`Jenkins-Crumb`) to validate the request [1].

1.2. Using "Trigger builds remotely" with an Authentication Token

Jenkins provides a built-in feature to trigger builds remotely using a predefined authentication token. This method is simpler to configure for basic remote triggering needs [3].

- **Configuration:** Within the Jenkins job configuration, enable the option "Trigger builds remotely (e.g., from scripts)" and define a unique **Authentication Token**.

- **URL Structure:** The job can then be triggered by sending an HTTP request to a URL structured as `JENKINS_URL/job/JOBNAME/build?token=TOKEN_NAME` .
- **Benefit:** This approach allows external systems to trigger builds without requiring full user credentials, provided they have the correct token [3].

1.3. Using Jenkins CLI

The Jenkins Command Line Interface (CLI) provides a powerful way to interact with Jenkins from a terminal or script. It supports various commands, including triggering builds [1].

- **Command Example:** To build a job, the command would be `java -jar jenkins-cli.jar -s JENKINS_URL build JOBNAME` .
- **Authentication:** The CLI can authenticate using SSH keys or API tokens, offering flexible security options [1].

1.4. Webhooks (e.g., GitHub, Generic Webhook Trigger)

Webhooks are an event-driven mechanism where Jenkins listens for specific events from external services (like source code management systems) and triggers builds accordingly.

- **GitHub Webhook:** The GitHub plugin for Jenkins allows jobs to be automatically triggered by events such as `push` events to a repository. This is a common practice for Continuous Integration (CI) workflows.
- **Generic Webhook Trigger Plugin:** For more generalized webhook integrations, the "Generic Webhook Trigger Plugin" allows Jenkins to trigger jobs based on any incoming HTTP POST request. It provides capabilities to parse JSON or XML payloads from the webhook and use that data in the build process [4].

2. Jenkins Installation Procedure

Installing Jenkins involves meeting certain prerequisites and following platform-specific steps. This section outlines the general requirements and installation guides for Ubuntu, Windows, and Docker.

2.1. Prerequisites

Before installing Jenkins, ensure your system meets the following requirements:

- **Java Development Kit (JDK):** Jenkins requires a compatible Java Runtime Environment (JRE) or JDK. As of recent Jenkins versions, **Java 17 or 21** is typically required [5]. It is crucial to install Java before Jenkins to avoid startup issues.
- **Hardware:**

- **Minimum:** 256 MB RAM, 1 GB drive space (10 GB recommended if running Jenkins as a Docker container).
- **Recommended for Small Teams:** 4 GB+ RAM, 50 GB+ drive space [6].

2.2. Installation on Ubuntu/Debian

For Debian-based distributions like Ubuntu, Jenkins can be installed using the `apt` package manager [5].

1. **Install Java:** First, update your package list and install OpenJDK 21:

Bash

```
sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
```

2. **Add Jenkins Repository Key:** Download and add the Jenkins GPG key to your system:

Bash

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
```

3. **Add Jenkins Repository to APT Sources:** Add the Jenkins stable repository to your system's `apt` sources list:

Bash

```
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

4. **Install Jenkins:** Update your package list again and install Jenkins:

Bash

```
sudo apt update
sudo apt install jenkins
```

5. **Start Jenkins Service:** Enable and start the Jenkins service:

Bash

```
sudo systemctl enable jenkins
sudo systemctl start jenkins
sudo systemctl status jenkins
```

Jenkins will typically listen on port 8080 [5].

2.3. Installation on Windows

The simplest way to install Jenkins on Windows is by using the official Windows MSI installer [7].

1. **Download Installer:** Obtain the latest .msi installer from the official Jenkins website's downloading page.
2. **Run Setup Wizard:** Execute the downloaded MSI file and follow the on-screen instructions of the Installation Setup Wizard.
3. **Service Logon Credentials:** During installation, you will be prompted to configure the service logon credentials. It is recommended to run Jenkins as an independent Windows service using a dedicated local or domain user account for enhanced security, rather than the LocalSystem account [7].
4. **Port Selection:** Specify the HTTP port Jenkins will use (default is 8080). You can test if the port is free during this step.
5. **Java Home Directory:** The installer will detect existing Java installations. Ensure it points to a compatible Java 17 or 21 JRE/JDK installation.
6. **Complete Installation:** Proceed with the installation. Jenkins will be installed as a Windows Service.

2.4. Installation via Docker

Running Jenkins in a Docker container is a popular method for its portability and isolation. This approach uses the official jenkins/jenkins Docker image [8].

1. **Pull Jenkins Image:** Download the Long-Term Support (LTS) Jenkins image from Docker Hub:

Bash

```
docker pull jenkins/jenkins:lts
```

2. **Run Jenkins Container:** Execute the following command to run Jenkins in a Docker container. This command maps port 8080 (Jenkins web UI) and 50000 (for build agents) and mounts a Docker volume for persistent data storage:

Bash

```
docker run -d -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home
```

- `-d` : Runs the container in detached mode.
- `-p 8080:8080` : Maps the container's port 8080 to the host's port 8080.
- `-p 50000:50000` : Maps the container's port 50000 to the host's port 50000, used for Jenkins agents.
- `-v jenkins_home:/var/jenkins_home` : Creates a named Docker volume `jenkins_home` and mounts it to `/var/jenkins_home` inside the container, ensuring data persistence.
- `--name jenkins` : Assigns the name `jenkins` to the container.
- `jenkins/jenkins:its` : Specifies the Docker image to use.

2.5. Post-Installation Setup (All Platforms)

After the initial installation, a web-based setup wizard guides you through the final configuration steps:

1. **Unlock Jenkins:** When accessing Jenkins for the first time (typically at `http://localhost:8080`), you will be prompted to unlock it using an automatically generated initial admin password. This password can be found in the Jenkins logs or in a file like `/var/jenkins_home/secrets/initialAdminPassword` (for Linux/Docker) [9].
2. **Install Plugins:** The wizard will suggest installing a set of recommended plugins. It is generally advisable to proceed with this option.
3. **Create First Administrator User:** Create your initial administrator account with a username and password.
4. **Configure Jenkins URL:** Confirm the Jenkins instance's root URL.